Borel degree-plus-one edge coloring from asymptotic separation index one

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Editorial note. Only the ASI1 degree-plus-one theorem is counted. The two ray/third-color lemmas are proved in full as solution obligations. No theorem establishing ASI1 for particular groups is assumed or counted.

Editorial note. For any ray, the argument applies to every tail, so a point in U0 may be followed by a later point outside its2d-neighborhood when checking the annulus crossings184.

Editorial note. The third color used for different generator pairs occurs only on edges whose endpoints lie in their pairwise disjoint V_i, which is the reason it creates no cross-generator color conflict.

Editorial scope. One Borel edge-coloring theorem with its complete local geometric/Borel construction. Borel maximal independent sets and finite-component choice are the explicit standard inputs. The finite-extension consequences and rank-specific abelian classification are outside this counted task.

Allowed standard prerequisites (editorial).

- Borel maximal independent sets in locally finite Borel graphs. Every locally finite Borel graph admits a Borel maximal independent set.. Kechris-Solecki-Todorcevic1999, Proposition4.2, explicitly invoked at193.. Applied to a locally finite line graph to obtain a maximal matching inside each selected ray-hitting region.
- Borel finite choices. Finite connected components and finitely many colorings per finite component permit a Borel choice, using a fixed Borel order.. Choose alternating two-edge-colorings of finite paths or even cycles; the global disjoint regions and edge cuts are not given.

Let X be a standard Borel space and $G \subset X \times X$ a Borel graph on X. An *edge coloring* of G is a function $c : G \to Y$ which sends adjacent edges in G, that is, edges sharing a vertex, to distinct elements of Y, or "colors". Such a c is called a *k-coloring* if $|Y| = k$. The *edge chromatic number* of G, denoted $\chi'(G)$, is the smallest cardinal k such that G admits a k-coloring. The *Borel edge-chromatic number* of G, denoted $\chi'_B(G)$, is the smallest cardinal k such that G admits a Borel k-coloring. See [1] for a survey of these and related notions.

The Borel graphs of interest to us in this paper will be those generated by Borel actions of finitely generated groups. A *marked group* is a pair (Γ, S) where Γ is a group and S is a finite symmetric generating set for Γ not containing 1.

We will sometimes omit the S if it will not cause confusion. Let X be a standard Borel space and $a : \Gamma \curvearrowright X$ a free Borel action. Let $G(a, S)$ denote the *Schreier graph* generated by this action. This is the graph defined by setting $x, y \in X$ adjacent if and only if $\gamma \cdot_a x = y$ for some $\gamma \in S$.

Note that all the notions in the above paragraph make sense even if S does not generate the entire group Γ. We shall sometimes use them in this case.

We start with the discrete versions of these notions: Let G be a locally finite graph on a set X, and N a positive integer. Let $G^{\leq N}$ denote the *distance N-graph* of G. This is the graph which sets two vertices adjacent if and only if there is a path from one to the other of length at most N.

The *asymptotic separation index* of G, denoted $\mathrm{asi}(G)$ is the infimum over $s \in \omega$ such that the following holds: For every positive integer N, X can be partitioned into $s+1$ sets, say $U_0, \ldots, U_s$, such that for each $0 \leq i \leq s$, the induced graph $G^{\leq N} \restriction U_i$ has only finite connected components. The *asymptotic dimension* of G, denoted $\mathrm{asdim}(G)$, is defined similarly, except now the connected components of each $G^{\leq N} \restriction U_i$ are required to have a uniform (finite) bound on their diameter.

Let G now be a Borel graph on a standard Borel space X. The *Borel asymptotic separation index* and *Borel asymptotic dimension* of G, denoted $\mathrm{asi}_B(G)$ and $\mathrm{asdim}_B(G)$ respectively, are defined exactly as their discrete counterparts were, except that now all the U_i's are required to be Borel sets.

Now let Γ be a finitely generated group, and $a : \Gamma \curvearrowright X$ a free Borel action. It is easy to verify that none of the above numbers for the graph $G(a, S)$ depend on the choice of generating set S for Γ. We can thus talk about the (Borel) asymptotic separation index and (Borel) asymptotic dimension of the action a itself.

Let G be a graph on a set X. An *injective G-ray* is an injective infinite sequence $x_0, x_1, \ldots \in X$ such that $(x_i, x_{i+1}) \in G$ for all $i \in \omega$. Given a set $U \subset X$ and an $n \in \omega$, let us write $B(U, n)$ for the ball of radius n around U. That is, the set of all points $x \in X$ whose path distance to U is less than or equal to n. For $m \in \omega$, let us write $A(U, n, m) = B(U, m) - B(U, n)$. One can check that if G and U are Borel, these sets are always Borel.

Theorem 1. *Let $a : \Gamma \curvearrowright X$ be a free Borel action of a marked group (Γ, S) on a standard Borel space X with Borel asymptotic separation index 1, and such that none of the elements of S have odd order. Then $\chi'_B(G(a, S)) \leq |S| + 1$.*

Lemma 1. *Suppose G is a locally finite Borel graph on a space X with $\mathrm{asi}_B(G) = 1$, and $d \in \omega$. Then we can find pairwise disjoint Borel sets $V_1, \ldots, V_d \subset X$ such that any injective G-ray contains infinitely many edges within each V_i.*

Proof. Since the tail of an injective G-ray is still an injective G-ray, it suffices to prove this with "infinitely many edges" replaced by "an edge".

Let $X = U_0 \sqcup U_1$ be a Borel partition of X witnessing $\mathrm{asi}_B(G) = 1$ for $N = 4d + 1$. That is, such that the graphs $G^{\leq N} \restriction U_0$ and $G^{\leq N} \restriction U_1$ both have only finite connected components. Set $V_i = A(U_0, 2(i-1), 2i)$ for each $1 \leq i \leq d$. Also set $W = X \setminus B(U_0, 2d) \subset U_1$. As was mentioned above, these are all still Borel. Let us now show these V_i's work.

Let $x = (x_n)$ be an injective G-ray. We first show that x contains some point from W. Suppose not. Then for each n, there is a $y_n \in U_0$ whose path distance to x_n is less than or equal to $2d$. Now for each n, going from y_n to

x_n, then to x_{n+1}, then to y_{n+1}, produces a path from y_n to y_{n+1} of length at most $2d + 1 + 2d = N$, so all the y_n's are in the same $G^{\leq N} \restriction U_0$-connected component. By definition of U_0, then, the set $\{y_n \mid n \in \omega\}$ is finite. But now x gives infinitely many points within distance $2d$ of this set, contradicting local finiteness.

Since $W \subset U_1$, the above shows in particular that x contains some point from U_1. Similarly, x must contain some point from U_0. Let $x_n \in W$ and $x_m \in U_0$. We can assume $m < n$. Now the path distance from x_i to U_0 changes by at most 1 each time i is incremented by 1. Thus, since x_m has distance 0 from U_0 and x_n has distance greater than $2d$ from U_0, for each $1 \leq i \leq d$, there must be some $m < j < n$ such that x_j and x_{j+1} have distance $2i - 1$ and $2i$ respectively from U_0. Then (x_j, x_{j+1}) is an edge of x contained in V_i, as desired. □

Lemma 2. *Let $a : \mathbb{Z} \curvearrowright X$ be a free Borel action of $\mathbb{Z}$, and S the usual generating set for $\mathbb{Z}$. Suppose $V \subset X$ is a Borel set with the property that every injective $G(a, S)$-ray contains infinitely many edges within V. Then there is a Borel 3-edge coloring of $G(a, S)$, say with the colors $1, 2$, and 3, such that the color 3 only occurs on edges contained within V.*

Proof. A well known theorem from [2] (Proposition 4.2) states that for locally finite Borel graphs, it is always possible to find Borel maximal independent sets. Apply this to the edge graph of $G(a, S) \restriction V$ to get a maximal set of edges A contained within V such that no two edges in A share a vertex. Give all the edges in A the color 3.

By maximality and the defining condition for the set V, these edges occur infinitely often in both directions along each $\mathbb{Z}$-orbit. Thus, the connected components of $G(a, S) \setminus A$ are all simply finite paths. These can of course be 2-colored with the colors 1 and 2, and since they are all finite, this can be done in a Borel fashion. (For example, by fixing a Borel linear order on the space of 2-edge colored finite subgraphs of $G(a, S)$, and picking the least 2-coloring in this order for each path above.) □

Proof. By hypothesis, we may write $S = S_0 \sqcup S_1$, where S_0 contains all of our generators of infinite order, and S_1 contains all of our elements of even (finite) order. For each $\gamma \in S_1$, the connected components of the graph $G(a, \{\gamma^{\pm 1}\})$ are all even (finite) cycles, or just single edges if γ has order 2. Thus they may be 2-colored in a Borel fashion as in the previous proof (or just 1-colored if γ has order 2). Thus we may Borel $|S_1|$-color the edges in $G(a, S_1)$.

It now suffices to Borel $|S_0|+1$-color the edges in $G(a, S_0)$ using a disjoint set of colors. Let $S_0 = \{\gamma_1^{\pm 1}, \ldots, \gamma_d^{\pm 1}\}$, so that $|S_0| = 2d$. We will use the color set $\{1, \ldots, 2d + 1\}$. Let $V_1, \ldots, V_d \subset X$ be the sets from Lemma 1. Since $\langle \gamma_i \rangle \cong \mathbb{Z}$ for each i, by Lemma 2 we may for each i color the edges in $G(a, \{\gamma_i^{\pm 1}\})$ using the colors $2i-1$, $2i$, and $2d+1$ such that the color $2d+1$ only appears on edges contained within V_i. Since the V_i's are pairwise disjoint, this does not cause any color conflicts, so we are done. □

References

[1] A.S Kechris and A.S. Marks, Descriptive graph combinatorics, preprint, 2019; posted in http://www.math.caltech.edu/∼kechris/

[2] A.S. Kechris, S. Solecki, and S. Todorcevic, Borel chromatic numbers, *Adv. Math.*, 141 (1999), 1-44.